

Low Level Language vs High Level Language

1. WHAT IS A PROGRAMMING LANGUAGE?

 A programming language is used to:

- Communicate with computer
 - Write instructions (programs)
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2. LOW LEVEL LANGUAGE

 Closest to hardware (machine)

Types:

- Machine Language (0s and 1s)
 - Assembly Language
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Features:

- ✓ Very fast execution
 - ✓ Direct communication with hardware
 - ✓ Efficient memory usage
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Disadvantages:

- ✗ Difficult to understand
 - ✗ Error-prone
 - ✗ Machine dependent
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Example:

10101010

3. HIGH LEVEL LANGUAGE

 Human-readable language

Examples:

- C
- Java
- Python

Features:

- ✓ Easy to read and write
- ✓ Easy debugging
- ✓ Portable (works on multiple systems)

Disadvantages:

- ✗ Slower than low-level
- ✗ Needs translator (compiler/interpreter)

Example:

```
printf("Hello");
```

4. KEY DIFFERENCE

Feature	Low Level Language	High Level Language
Readability	Very difficult	Easy
Speed	Very fast	Slower
User Friendly	No	Yes
Dependency	Machine dependent	Portable